

Parcel Promise Guard

Proactive SLA monitoring and remediation platform for parcel logistics

Problem Statement

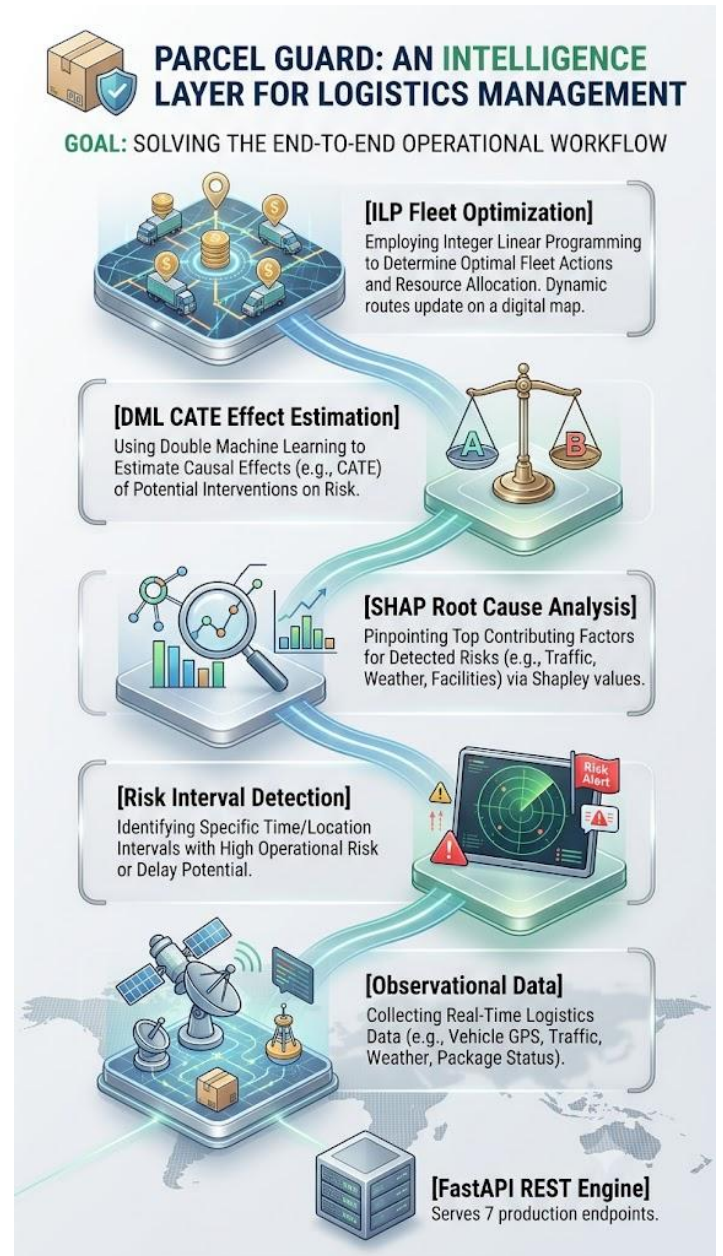
Context & Business Domain

Modern e-commerce and third-party logistics (3PL) networks process millions of time-critical shipments daily. Maintaining delivery reliability and adhering to promised Service Level Agreements (SLAs) is essential for customer retention, operational cost control, and brand equity.

However, existing supply chain execution platforms operate **reactively**. SLA breaches are typically identified only when a parcel fails to arrive at a distribution hub or when the delivery window has already elapsed. This reactive approach leads to customer dissatisfaction, brand churn, and millions of dollars in contractual SLA breach penalties.

The goal of **Parcel Guard** is to build an overlay intelligence layer for logistics management that solves the complete end-to-end operational workflow:

Observational Data → Risk Interval Detection → SHAP Root Cause
→ DML CATE Effect → ILP Fleet Optimization



System Architecture

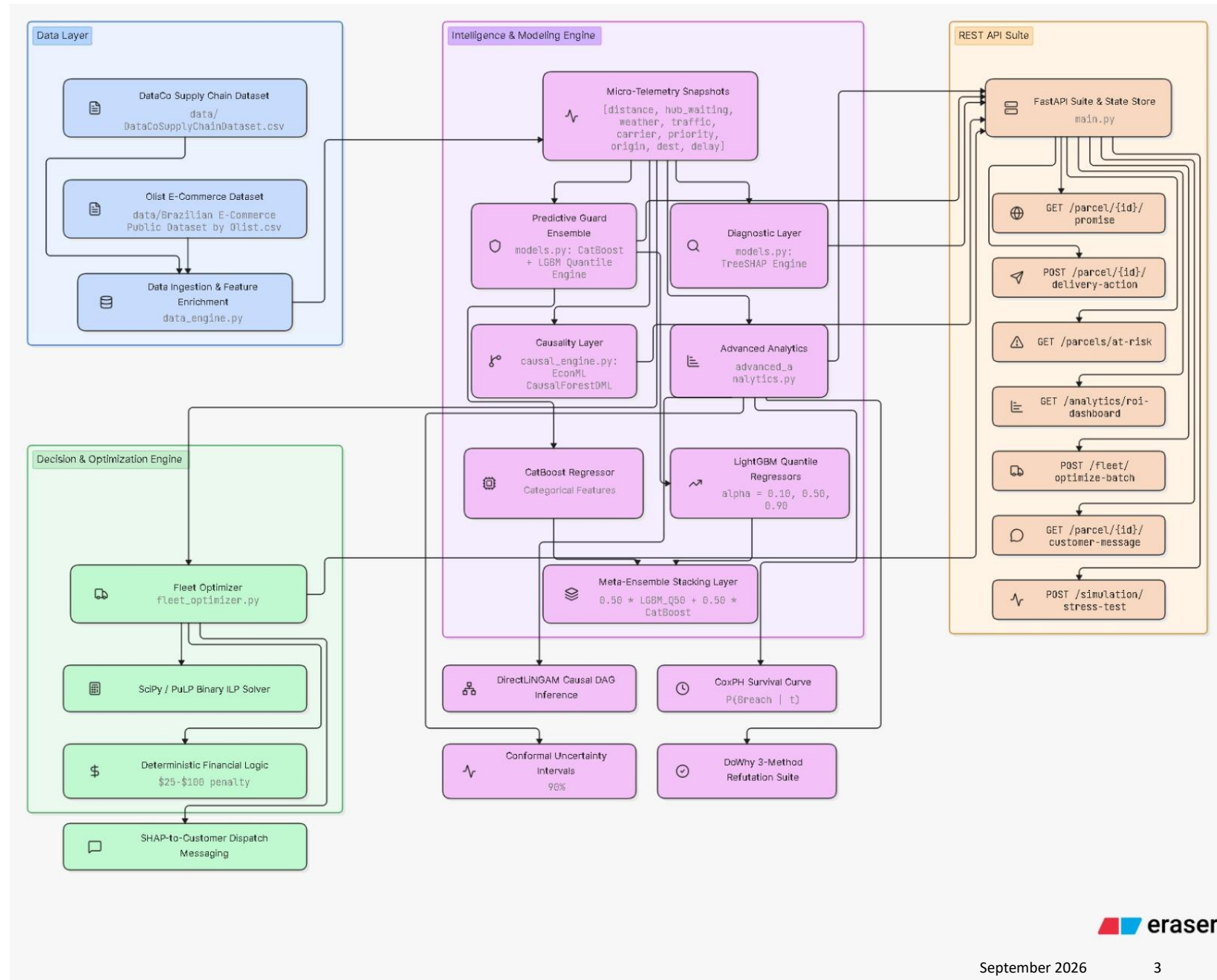
Parcel Guard monitors proactively for SLA breaches :

Detects impending SLA breaches hours in advance using uncertainty-aware quantile ML ensembles.

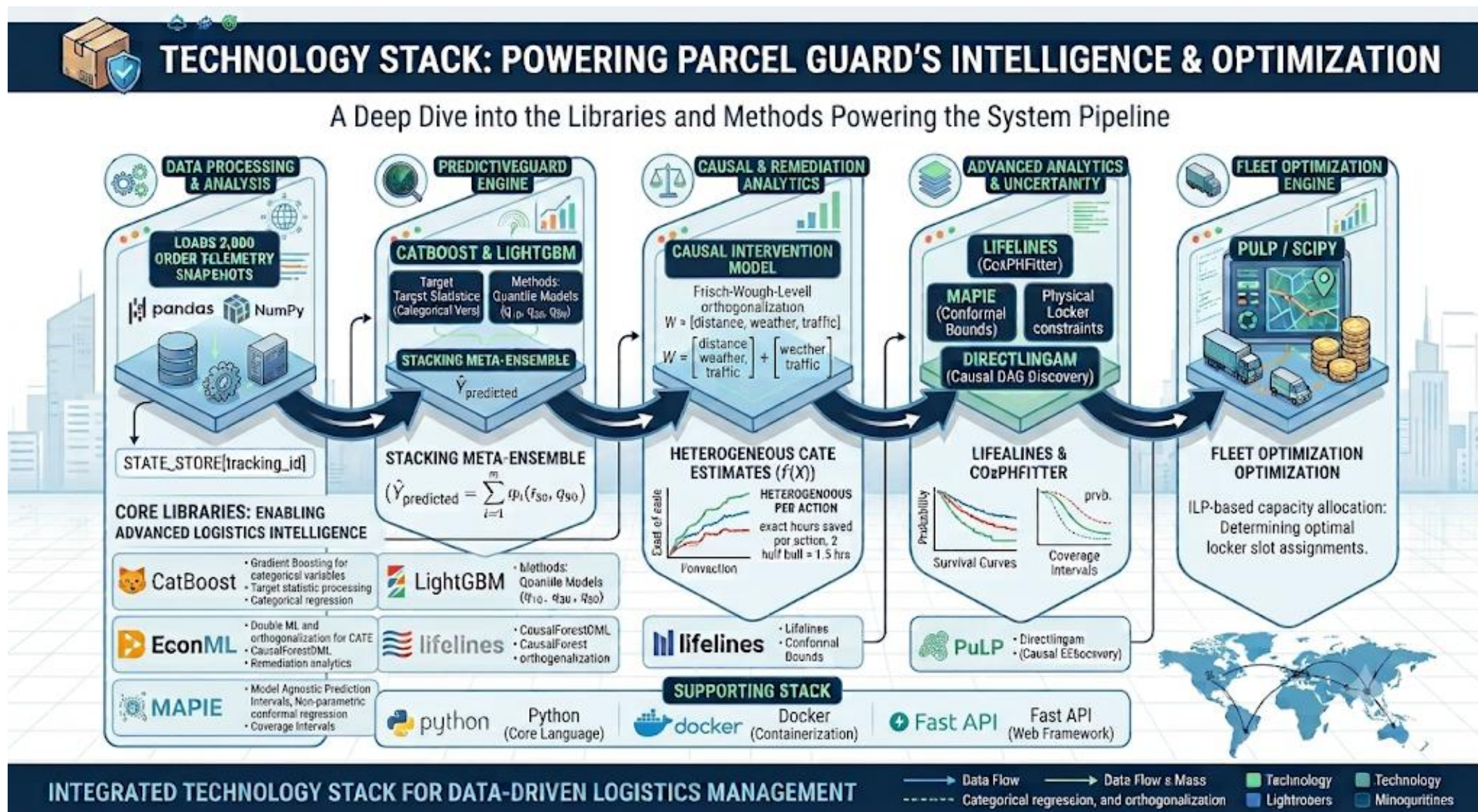
Diagnoses the exact root causes (e.g., severe weather vs sorting hub dwell time) using game-theoretic TreeSHAP.

Recommends Causal Interventions (e.g., rerouting to a parcel locker) by calculating unconfounded heterogeneous treatment effects (CATE $\tau(X)$) via Double Machine Learning.

Optimizes Fleet Capacity using Integer Linear Programming (ILP) to maximize net financial dollars saved under physical locker constraints.



Technology used :



Dataset creation

- **DataCo Supply Chain Dataset (DataCoSupplyChainDataset.csv)**: Extracts real order IDs (Order Id) and shipping dates (shipping date (DateOrders)) , Formats tracking IDs as DC-<order_id> .
Olist E-Commerce Dataset (Brazilian E-Commerce Public Dataset by Olist.csv): Extracts purchase timestamps (order_purchase_timestamp) and order IDs (order_id) , Formats tracking IDs as OL-<order_id>. **Fallback Mechanism**: If local files are not present or sample sizes are under the target n_samples, synthetic fallback tracking IDs (ORD-90000+i) and timestamps are generated.
- **Synthetic Data Generation**: In real logistics, rerouting decisions ($T = \text{locker_rerouted}$) are not assigned randomly—packages experiencing severe weather or long remaining distances are more likely to be rerouted by dispatchers. To simulate observational confounding:
- **Confounders W** : [distance_remaining_km, weather_severity, traffic_index]
- **Propensity Score Equation**: $\text{logit}(W) = 0.005 \cdot \text{distance_remaining_km} + 2.0 \cdot \text{weather_severity} - 1.5$ $e(W) = P(\text{locker_rerouted} = 1 \mid W) = \frac{1}{1 + \exp(-\text{logit}(W))}$
- **Treatment Assignment**: $\text{locker_rerouted} \sim \text{Bernoulli}(e(W))$
- **Heterogeneous Treatment Effect (CATE Mechanism $\tau(X)$)**
- The effect of rerouting a parcel to a locker depends heterogeneously on hub waiting time:
- **Severe Hub Congestion (hub_waiting_time_hrs > 1.5 hrs)**: Rerouting bypasses the bottleneck, saving **3.0 hours** ($\tau(X) = -3.0$).
- **Normal Flow (hub_waiting_time_hrs ≤ 1.5 hrs)**: Rerouting provides a smaller benefit of **0.5 hours** ($\tau(X) = -0.5$).
- $$\tau(X) = \begin{cases} -3.0 \text{ hours,} & \text{if } \text{hub_waiting_time_hrs} > 1.5 \\ -0.5 \text{ hours,} & \text{otherwise} \end{cases}$$
- The final actual delay hours (final_actual_delay_hrs) are calculated using a Data Generating Process (DGP) structural equation with additive Gaussian noise $\epsilon \sim \mathcal{N}(0, 0.5^2)$:
- $$Y = 0.6 \cdot \text{hub_waiting_time_hrs} + 3.0 \cdot \text{weather_severity} + 2.5 \cdot \text{traffic_index} + \tau(X) \cdot \text{locker_rerouted} + \epsilon$$
-

Application flow

1. *Application startup :*

- - **Loads 2,000 Order Telemetry Snapshots:** Combines real order metadata from DataCo and Olist datasets with dynamic micro-telemetry (distance, weather, traffic, carrier, priority).
- - **Trains ML Models On Startup:**
 - PredictiveGuard: Trains CatBoost + LightGBM Quantile models.
 - CausalRemediationEngine: Fits EconML CausalForestDML.
 - AdvancedCausalAnalytics: Fits CoxPHFitter survival model and conformal calibration residual bounds.
- **Populates In-Memory State Store:** Loads active parcel records into STATE_STORE[tracking_id].
- **2. SLA Breach Detection & Risk Engine:** When a request comes in (e.g. *GET /parcel/{tracking_id}/promise*), PredictiveGuard evaluates the parcel:
- **CatBoost Regressor:** Processes categorical variables (carrier_id, priority_tier, origin_hub, destination_hub) using target statistics.
- **LightGBM Quantile Regressors:** Predicts 3 quantiles:
 - q_{10} (delay_lower_bound_hrs): Optimistic delay bound.
 - q_{50} (delay_median_hrs): Median delay.
 - q_{90} (delay_upper_bound_hrs): Conservative upper risk bound.
- **Stacking Meta-Ensemble:** Combines q_{50} and CatBoost predictions: $\hat{Y}_{predicted} = 0.50 \cdot \hat{q}_{0.50} + 0.50 \cdot \hat{Y}_{CatBoost}$
- **Risk Flagging:** If $q_{90} > 0$, an SLA breach risk is flagged immediately, even if median prediction is near zero.
- **3. Root Cause Diagnostics :**
 - TreeSHAP calculates exact additive Shapley feature attributions: $\hat{Y}(x) = \phi_0 + \sum_i \phi_i(x)$
 - It extracts the **top 3 positive delay drivers** (e.g., weather_severity added +1.63 hours delay, hub_waiting_time_hrs added +0.58 hours delay).

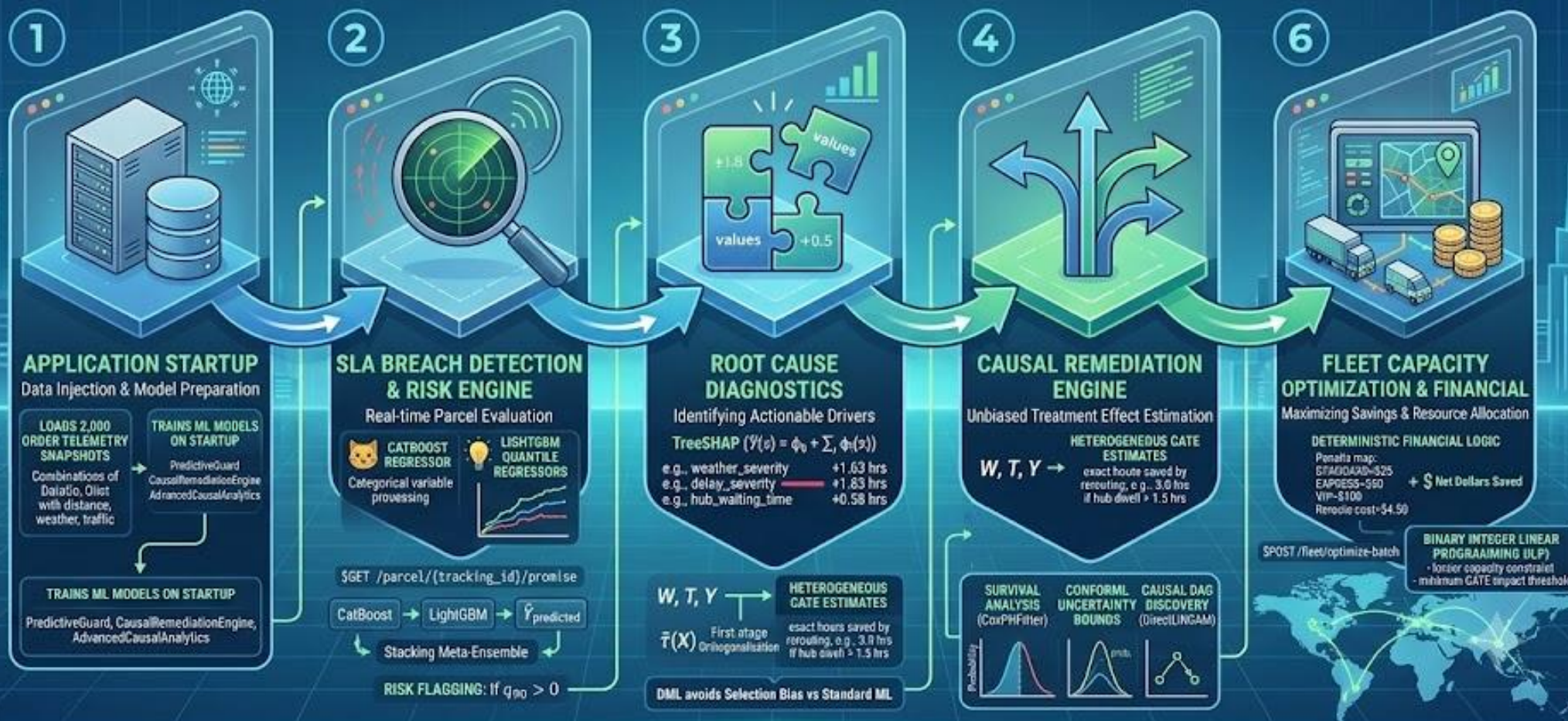
APPLICATION FLOW CONTINUED

- **4. Causal Remediation Engine (*causal_engine.py*)**
 - Standard ML models suffer from selection bias (e.g., operational staff reroute packages mostly during severe weather, creating confounding).
 - **Double Machine Learning (DML)** uses Frisch-Waugh-Lovell orthogonalization:
 - First-stage LightGBM models strip away confounder effects $W = [distance, weather, traffic]$ from both treatment T (locker_rerouted) and outcome Y (delay).
 - Estimates Heterogeneous CATE $\hat{\tau}(X)$: exact hours saved by rerouting this specific parcel to a locker (e.g., saves 3.0 hrs if hub dwell > 1.5 hrs, saves 0.5 hrs otherwise).
- **5. Advanced Analytics (*advanced_analytics.py*)**
 - **Survival Analysis (lifelines.CoxPHFitter)**: Computes dynamic hazard curves $h(t | X)$ and continuous breach probabilities $P(Breach | N \text{ hrs})$ at $N \in \{2, 6, 12, 24\}$ hours.
 - **Conformal Uncertainty Bounds (MAPIE)**: Computes non-parametric 90% coverage prediction intervals $[lower, upper]$ without making Gaussian distribution assumptions.
 - **Causal DAG Discovery (DirectLiNGAM)**: Empirically learns directed causal graphs between telemetry variables.
- **6. Fleet Capacity Optimization & Financial ROI (*fleet_optimizer.py*)**
 - When dispatchers run batch allocations (*POST /fleet/optimize-batch*):
 - **Deterministic Financial Logic**:
 - - Penalty map: STANDARD = \$25, EXPRESS = \$50, VIP = \$100.
 - - Reroute cost: \$4.50.
 - - Net Dollars Saved =
$$\begin{cases} \text{penalty_usd} - 4.50, & \text{if SLA breach is predicted and } \hat{\tau}(X) \geq 1.5 \text{ hrs} \\ 0.0, & \text{otherwise} \end{cases}$$
 - **Binary Integer Linear Programming (ILP)** via PuLP / SciPy: $\max_{x_{i,j}} \sum_{i=1}^N \sum_{j=1}^K (\text{penalty}_i - 4.50) \cdot x_{i,j}$ subject to $\sum_i x_{i,j} \leq C_j$ (locker capacity constraint) and $\hat{\tau}_i(X) \geq 1.5 \text{ hrs}$ (minimum CATE impact threshold)

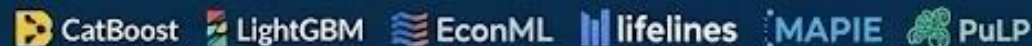


PARCEL GUARD: AN INTELLIGENCE LAYER FOR LOGISTICS MANAGEMENT

SYSTEM PIPELINE: ADVANCED LOGISTICS PREDICTION & OPTIMIZATION



INTEGRATED PLATFORM FOR DATA-DRIVEN LOGISTICS MANAGEMENT



BACKEND APIS



PARCEL GUARD: AN INTELLIGENCE LAYER FOR LOGISTICS MANAGEMENT

GOAL: SOLVING THE END-TO-END OPERATIONAL WORKFLOW

Endpoint	Method	Purpose
GET /parcel/{tracking_id}/promise	GET	Returns realtime delay prediction, risk bounds, SHAP root causes, CATE action, conformal intervals, survival probabilities, and financial ROI.
POST /parcel/{tracking_id}/delivery-action	POST	Executes parcel reroute to a locker (locker_rerouted = 1), updating state store and recalculating updated delay.
GET /parcels/at-risk	GET	Returns all active shipments predicted to breach SLA with at least min_lead_hours notice.
GET /analytics/roi-dashboard	GET	Aggregates fleet-wide metrics: total SLAs protected, net dollars saved, 6-hour Advance Detection Rate (ADR_6), and AUUC uplift score.
POST /fleet/optimize-batch	POST	Executes Integer Linear Programming (ILP) fleet allocation over at-risk parcels given physical locker slot constraints.
GET /parcel/{tracking_id}/customer-message	GET	Generates deterministic, SHAP-mapped SMS/push customer notifications.
POST /simulation/stress-test	POST	Simulates regional weather/traffic spikes, mutates parcel state, and triggers auto re-optimization.